



MySQL - RDBMS

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SELECT – DQL – WHERE

- It is always good idea to fetch only required rows (to reduce network traffic).
- The WHERE clause is used to specify the condition, which records to be fetched.
- Relational operators
 - <, >, <=, >=, =, != or <>
- NULL related operators
 - NULL is special value and cannot be compared using relational operators.
 - IS NULL or <=>, IS NOT NULL.
- Logical operators
 - AND, OR, NOT



SELECT – DQL – WHERE

- BETWEEN operator (include both ends)
 - c1 BETWEEN val1 AND val2
- IN operator (equality check with multiple values)
 - c1 IN (val1, val2, val3)
- LIKE operator (similar strings)
 - c1 LIKE 'pattern'.
 - % represent any number of any characters.
 - _ represent any single character.



UPDATE – DML

- To change one or more rows in a table.
- Update row(s) single column.
 - UPDATE table SET c2=new-value WHERE c1=some-value;
- Update multiple columns.
 - UPDATE table SET c2=new-value, c3=new-value WHERE c1=some-value;
- Update all rows single column.
 - UPDATE table SET c2=new-value;



DELETE vs TRUNCATE vs DROP

- **DELETE:** DML operation (can be rolled back)
- Used to remove one or more rows from a table.
- Syntax:
 - DELETE FROM table WHERE c1 = value;
- Deletes specific rows that match the condition.
- Example:
 - DELETE FROM employees WHERE id = 10;
- Delete all rows:
- DELETE FROM temp;
- Removes all rows but keeps table structure.

- **TRUNCATE:** DDL operation (cannot be rolled back)
- Used to delete all rows quickly.
- Syntax:
 - TRUNCATE TABLE table;
- Faster than DELETE.
- Cannot use WHERE with TRUNCATE.

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- **DROP :** DDL operation (cannot be rolled back)
 - Used to remove the table (and its structure) completely.
 - Syntax:
 - DROP TABLE table;
 - Deletes both data and table definition.
 - Optional (avoid errors):
 - DROP TABLE IF EXISTS table;
 - Delete a whole database/schema:
 - DROP DATABASE db;



Seeking HELP

- HELP is client command to seek help on commands/functions.
 - HELP SELECT;
 - HELP Functions;
 - HELP SIGN;



SQL functions

- RDBMS provides many built-in functions to process the data.
- These functions can be classified as:
 - Single row functions
 - One row input produce one row output.
 - e.g. ABS(), CONCAT(), IFNULL(), ...
 - Multi-row or Group functions
 - Values from multiple rows are aggregated to single value.
 - e.g. SUM(), MIN(), MAX(), ...
- These functions can also be categorized based on data types or usage.
 - Numeric functions
 - String functions
 - Date and Time functions
 - Control flow functions
 - Information functions
 - Miscellaneous functions



Numeric & String functions

- ABS()
- POWER()
- ROUND(), FLOOR(), CEIL()

- ASCII(), CHAR()
- CONCAT()
- SUBSTRING()
- LOWER(), UPPER()
- TRIM(), LTRIM(), RTRIM()
- LPAD(), RPAD()



Date-Time and Information functions

- VERSION()
- USER(), DATABASE()
- MySQL supports multiple date time related data types
 - DATE (3), TIME (3), DATETIME (5), TIMESTAMP (4), YEAR (1)
- SYSDATE(), NOW()
- DATE(), TIME()
- DAYOFMONTH(), MONTH(), YEAR(), HOUR(), MINUTE(), SECOND(), ...
- DATEDIFF(), DATE_ADD(), TIMEDIFF()
- MAKEDATE(), MAKETIME()



Control and NULL and List functions

- NULL is special value in RDBMS that represents absence of value in that column.
- NULL values do not work with relational operators and need to use special operators.
- Most of functions return NULL if NULL value is passed as one of its argument.
- ISNULL()
- IFNULL()
- NULLIF()
- COALESCE()

- GREATEST(), LEAST()

- IF(condition, true-value, false-value)



THANK YOU

